3D Organoid Feature Reference

Per-cell features extracted by 02-extract_features.py

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Overview

Each row in the output CSV corresponds to one segmented cell that passed quality filtering. Features are grouped into six categories: metadata, spatial coordinates, morphology, nuclear intensity, spatial topology, and 3D shape descriptors. All physical measurements are in micrometres (µm) or cubic micrometres (µm³) unless otherwise stated.

Total features per cell

- 36 features -- standard run (COMPUTE_DISTANCE_TO_BORDER = False)
- 38 features -- with border distance enabled (COMPUTE_DISTANCE_TO_BORDER = True)

Primary input arrays

- cell_masks_filtered -- integer label mask for cell bodies (one label per cell)
- nuclei_masks_filtered -- integer label mask for nuclei (matching label IDs)
- Raw OME-TIFF -- DAPI channel (channel 0) used for nuclear intensity features

Output

- One CSV file per organoid written to output/features/<stem>_features.csv
- Run after segment_organoids.py. Feature parameters can be changed and the script re-run without repeating segmentation.

Configuration Parameters

Edit the CONFIGURATION block at the top of 02-extract_features.py. These settings affect which files are processed and how features are computed.

Parameter	Default	Description
PROCESS_STEMS	None	List of filename prefixes to process. None = process all organoids found in output/segmentation/.
BATCH	"Batch1"	Batch label written to every CSV row. Groups organoids from one experimental session.
SAMPLE	"PDAC_osmotic"	Sample or cell-line label written to every CSV row.
WELL	"A01"	Well identifier written to every CSV row. Matches plate-reader notation.
FIELD	"s1"	Field-of-view identifier written to every CSV row.
READ_VOXEL_FROM_OME	True	Read XY and Z voxel size from OME-XML metadata. Set to False and fill PIXEL_SIZE_XY / PIXEL_SIZE_Z manually if OME metadata is absent.
PIXEL_SIZE_XY	0.414	Fallback XY pixel size in µm. Used only when READ_VOXEL_FROM_OME = False.

Parameter	Default	Description
PIXEL_SIZE_Z	1.0	Fallback Z step size in µm. Used only when READ_VOXEL_FROM_OME = False.
RADIUS_UM	20.0	Neighbourhood radius in µm for topology features. All cells within this sphere are counted as neighbours. Use 20 µm for PDAC (cell radius ~8 µm); use 35–40 µm for larger cells such as HMECyst (~13 µm radius).
COMPUTE_DISTANCE_TO_BORDER	True	Compute nuclei_distance_to_border_um and nuclei_distance_to_border_ratio. Uses one distance_transform_edt call per organoid. Set to False to skip and reduce runtime.
OVERWRITE	True	Overwrite existing feature CSVs. Set to False to skip organoids that already have a CSV.

1. Metadata

Identifier columns written from the configuration block. All five values are identical for every cell in one batch run. They are used to merge CSVs from multiple organoids into a single analysis dataframe.

Feature name	Type	Unit	Description
Batch	string	—	Batch identifier from config. Groups multiple organoids from one experimental session.
Sample	string	—	Sample or cell-line label from config (e.g. PDAC, HMECyst).
Well	string	—	Well identifier from config (e.g. A01). Matches plate-reader notation.
Field	string	—	Field-of-view identifier from config (e.g. s1). Useful when multiple images come from one well.
Cell_Number	int	—	Label index of this cell in the filtered mask. Unique within one image. Matches the integer value in cell_masks_filtered.tif.

2. Spatial Coordinates

Physical position of each cell centroid in three coordinate systems: absolute image coordinates, organoid-centred relative coordinates, and a normalised radial position. All three are provided so the same data can be used for image registration (absolute), organoid-level analysis (centred), and radial zone analysis (normalised).

Note: *The organoid centre of mass is computed as the mean centroid of all cells that passed quality filtering. The organoid radius is defined as the 95th percentile of all centroid-to-centre distances.*

Feature name	Type	Unit	Description
centroid_z_um	float	µm	Z position of the cell centroid measured from the top of the volume. Increases with imaging depth.
centroid_y_um	float	µm	Y position of the cell centroid from the top edge of the image.
centroid_x_um	float	µm	X position of the cell centroid from the left edge of the image.
centroid_z_rel_um	float	µm	Z position relative to the organoid centre of mass. Negative = above centre, positive = below.
centroid_y_rel_um	float	µm	Y position relative to the organoid centre of mass.

Feature name	Type	Unit	Description
centroid_x_rel_um	float	µm	X position relative to the organoid centre of mass.
radial_dist_um	float	µm	Euclidean distance from the organoid centre of mass to this cell centroid. Direct measure of depth within the organoid.
radial_dist_norm	float	—	radial_dist_um divided by the organoid radius (95th percentile of all centroid distances). 0 = organoid centre, 1 = surface. Values slightly above 1.0 are normal for outermost cells.
zone	string	—	Discrete radial zone label derived from radial_dist_norm. core: < 0.50. intermediate: 0.50 to 0.75. periphery: > 0.75.

3. Cell and Nuclear Morphology

Size and shape of segmented cell bodies and their nuclei. Volume features are computed directly from voxel counts. Shape features use two approaches: a 2D proxy from skimage regionprops (consistent and fast) and true 3D principal-axis descriptors from PCA on voxel coordinates (more accurate for elongation; see Section 6).

3.1 Volumes

Feature name	Type	Unit	Description
cell_volume_um3	float	μm^3	Volume of the cell body in cubic micrometres. Computed as voxel count × (pixel_xy² × pixel_z).
nuclei_volume_um3	float	μm^3	Volume of the nucleus in cubic micrometres. Same calculation applied to the nucleus label mask.

3.2 Distance to organoid border (optional)

Only present when COMPUTE_DISTANCE_TO_BORDER = True. Computed from a single distance_transform_edt on the organoid binary mask -- one transform for the whole organoid, not per cell. This is much faster than per-cell computation and gives equivalent results.

Feature name	Type	Unit	Description
nuclei_distance_to_border_um	float	µm	Distance from the nucleus centroid to the nearest organoid surface voxel. High value = deep interior, low value = near surface.
nuclei_distance_to_border_ratio	float	—	nuclei_distance_to_border_um divided by the maximum border distance within the same cell (approximate cell radius). 0 = nucleus at the cell edge, 1 = nucleus at the cell centre. Used in Ong et al. (2025) as a nuclear polarity measure.

3.3 Shape (regionprops proxy)

axis_major_length and axis_minor_length are computed by skimage regionprops on the 3D label volume and report the axes of the largest 2D cross-section. They are a consistent proxy for elongation but are not true 3D axes. True 3D axes are in Section 6.

Feature name	Type	Unit	Description
cell_elongation	float	—	Major axis / minor axis. 1.0 = round, higher = more elongated. Equivalent to the major/minor ratio used in Ong et al. (2025).

Feature name	Type	Unit	Description
cell_roundedness	float	—	Minor axis / major axis. Inverse of cell_elongation. 1.0 = round, lower = more elongated. Range: 0 to 1.

4. Nuclear Intensity (DAPI Channel)

Measured from the raw (pre-normalisation) DAPI array at all voxels belonging to the nucleus label mask. These features capture chromatin organisation, staining quality, and the effects of nuclear condensation or dissolution. They are measured on the matched nucleus only -- not the entire cell body.

Feature name	Type	Unit	Description
CV_chromatin	float	—	Coefficient of variation of DAPI intensity: std / mean. High CV = spatially heterogeneous chromatin (open, active). Low CV = uniform, compacted chromatin. Primary discriminator for osmotic stress conditions.
max_intensity_nuclear	float	AU	Peak DAPI intensity voxel inside the nucleus. Sensitive to bright chromatin foci and pyknotic condensation in necrotic cores.
min_intensity_nuclear	float	AU	Minimum DAPI intensity voxel inside the nucleus. Low values indicate dim interior regions or incomplete staining penetration.
avg_intensity_nuclear	float	AU	Mean DAPI intensity inside the nucleus. Reflects average chromatin density and staining depth penetration.
std_intensity_nuclear	float	AU	Standard deviation of DAPI intensity inside the nucleus. Higher = more spatially heterogeneous chromatin distribution.
sum_intensity_nuclear	float	AU	Total integrated DAPI signal in the nucleus. Proportional to total DNA content × staining penetration depth.

Note: *AU = arbitrary units. Intensity values are taken from the raw 16-bit DAPI array before any normalisation. Comparisons between organoids are only valid when acquisition settings (laser power, gain, exposure) are identical across all images.*

5. Spatial Topology

Neighbourhood relationships computed within a sphere of radius RADIUS_UM (default 20 µm) centred on each cell centroid. A KDTree is built from all cell centroids for efficient neighbourhood queries. These features characterise tissue packing architecture and are the primary discriminators between solid and cyst-forming epithelium.

Note: *All topology features depend on RADIUS_UM. Results are not directly comparable across datasets unless the same radius was used. For large cells (HMECyst, ~13 µm radius) use RADIUS_UM = 35–40 µm. For PDAC cells (~8 µm radius) use 20 µm.*

Feature name	Type	Unit	Description
n_nuclei_neighbors	int	—	Number of other cell centroids within RADIUS_UM of this cell. Direct count of local neighbours.
nb_nuclei_neighbors_ripley	float	—	n_nuclei_neighbors cast to float. Used as the Ripley-B input value in topology analysis pipelines.
distance_to_neighbors_mean_um	float	µm	Mean distance from this cell to all neighbours within RADIUS_UM. Lower = more tightly packed neighbourhood.
distance_to_neighbors_min_um	float	µm	Distance to the nearest neighbour. Approximates the cell-to-cell gap or contact distance.

Feature name	Type	Unit	Description
distance_to_neighbors_max_um	float	µm	Distance to the farthest neighbour within RADIUS_UM. Reflects neighbourhood spread.
local_density_per_mm3	float	/mm³	Number of neighbours divided by the search sphere volume in mm³. Standard nuclear density measure. Reference: HMECyst control = 94,869/mm³; HMECyst cyst = 120,329/mm³ (Ong et al. 2025).
crystal_distance_um	float	µm	Theoretical nearest-neighbour distance if cells were arranged in a perfect crystal at the observed local density. Computed as (sphere_volume / n_neighbours)^(1/3). Smaller = more tightly and regularly packed. Key discriminator for cyst vs solid organoid architecture.

6. 3D Shape Descriptors (Principal Axes)

True 3D shape descriptors computed from the eigenvalues of the voxel coordinate covariance matrix within each cell's bounding box. Voxel coordinates are corrected for anisotropy before PCA. The three principal axes correspond to the longest, intermediate, and shortest dimensions of the 3D cell extent. These are more accurate than the regionprops proxy axes in Section 3.3 for genuinely 3D objects.

Note: *Axis lengths are computed as 2 × sqrt(eigenvalue), analogous to the diameter of the best-fit ellipsoid along each principal axis.*

Feature name	Type	Unit	Description
major_axis_um	float	µm	Length of the longest principal axis. Corresponds to the cell's longest dimension in 3D.
medium_axis_um	float	µm	Length of the intermediate principal axis. Together with major and minor, describes the 3D ellipsoid that best fits the cell.
minor_axis_um	float	µm	Length of the shortest principal axis. Small relative to major = elongated or flat cell.
prolate_ratio	float	—	major_axis / medium_axis. High value = rod-like or spindle-shaped cell (prolate ellipsoid). Value near 1 = round or disc-like.
oblate_ratio	float	—	medium_axis / minor_axis. High value = disc-like or flattened cell (oblate ellipsoid). Value near 1 = round or rod-like. Used together with prolate_ratio to classify cell shape (sphere / rod / disc).

7. Analysis Tips

Osmotic stress (isotonic vs hypertonic PDAC)

The primary discriminating feature is CV_chromatin. Hypertonic stress causes chromatin compaction, reducing the coefficient of variation of DAPI intensity (Cohen's d ≈ 0.66 in Ong et al. 2025). Cell volume shows a slight decrease. Shape features (cell_elongation, cell_roundedness) and spatial topology features do not differ significantly between conditions. When analysing at the organoid level (per-organoid medians as the statistical unit), at least 5 organoids per condition are needed to reach adequate power for CV_chromatin.

Tissue architecture (HMECyst control vs cyst)

Topology features cleanly discriminate solid from cyst-forming organoids. The cyst organoid has more neighbours within the search radius, a smaller crystal_distance_um (17.3 vs 22.7 µm), and a higher Ripley ratio reflecting crystal-like regular spacing of cells in a shell monolayer. Use radial_dist_norm to confirm shell architecture: cyst cells cluster near 1.0 with no cells below radial_dist_norm = 0.78. Logistic regression and Random Forest classifiers both achieve AUC = 1.00 on topology features alone at the organoid level.

Radial subpopulation analysis (large PDAC organoid)

Use zone or radial_dist_norm to bin cells into radial bands. Expect a continuous gradient rather than abrupt clusters: core cells are smaller (cell_volume_um3), rounder (prolate_ratio near 1), and show higher intensity variation where necrosis is present (std_intensity_nuclear, CV_chromatin). Peripheral cells are larger and more elongated. k-means clustering (k=3, selected by silhouette analysis) produces a silhouette score around 0.17–0.19, confirming a continuum rather than discrete states. Intensity and size features dominate cluster separation; topology features contribute secondary effects.

Spatial visualisation

centroid_z_rel_um, centroid_y_rel_um, centroid_x_rel_um can be used directly as axes in a 3D scatter plot coloured by any feature. radial_dist_norm on the x-axis with any morphometric feature on the y-axis gives a radial profile plot showing how that feature changes from centre to surface. This is the recommended first plot for any new organoid type.

Quality control

Cells are filtered before feature extraction by volume range (removing debris and merged objects) and by nucleus-to-cell volume ratio. Check cell_volume_um3 and nuclei_volume_um3 distributions for each condition before analysis. Outliers at the extremes typically reflect segmentation errors rather than genuine biology.